

Part A: Numerical Expressions and Order of Operations

Evaluate each expression completely. Show each major order-of-operations step.

1. $11 + 6 \cdot 9$

6. $(-7 + 9)^2 \div 1$

2. $12 + (-3) \cdot 2$

7. $19 - 8 \cdot ((-6) + 19)$

3. $12 + (-3) \cdot 8$

8. $9^2 - 2 \cdot 2$

4. $-3 + (-2) \cdot (-3)$

9. $(-9 + 18)^2 \div (-9)$

5. $-11 - 7 \cdot (2 + (-11))$

10. $(3 + 9)^2 \div (-1)$

Part B: Properties of Addition and Multiplication

Select the property that is displayed in the given equation.

1. $8 + 4 = 4 + 8$

- A) Commutative Property of Addition
- B) Inverse Property of Addition
- C) Associative Property of Addition
- D) Identity Property of Addition

2. $4 \cdot 5 = 5 \cdot 4$

- A) Associative Property of Multiplication
- B) Identity Property of Multiplication
- C) Commutative Property of Multiplication
- D) Inverse Property of Multiplication

3. $(3 \cdot 5) \cdot 4 = 3 \cdot (5 \cdot 4)$

- A) Commutative Property of Multiplication
- B) Identity Property of Multiplication
- C) Associative Property of Multiplication
- D) Inverse Property of Multiplication

4. $(4 + 7) + 5 = 4 + (7 + 5)$

- A) Commutative Property of Addition
- B) Identity Property of Addition
- C) Inverse Property of Addition
- D) Associative Property of Addition

5. $0 + x = x$

- A) Associative Property of Addition
B) Inverse Property of Addition
C) Commutative Property of Addition
D) Identity Property of Addition
6. $1 \cdot a = a$
A) Inverse Property of Multiplication
B) Associative Property of Multiplication
C) Commutative Property of Multiplication
D) Identity Property of Multiplication
7. $2 \cdot \frac{1}{2} = 1$
A) Commutative Property of Multiplication
B) Inverse Property of Multiplication
C) Identity Property of Multiplication
D) Associative Property of Multiplication
8. $x \cdot \frac{1}{x} = 1$
A) Identity Property of Multiplication
B) Inverse Property of Multiplication
C) Associative Property of Multiplication
D) Commutative Property of Multiplication
9. $(a + 14) + 23 = a + (14 + 23)$
A) Commutative Property of Addition
B) Identity Property of Addition
C) Associative Property of Addition
D) Inverse Property of Addition
10. $(20 \cdot y) \cdot (-11) = 20 \cdot (y \cdot (-11))$
A) Commutative Property of Multiplication
B) Inverse Property of Multiplication
C) Associative Property of Multiplication
D) Identity Property of Multiplication

Part C: Evaluating Expressions by Substitution

Substitute the given value or values and evaluate completely.

- | | |
|--|---|
| 1. $-3c - c - 4p^2 - p$, $c = -3$, $p = 3$ | 3. $4a + 4a + 2j^2 + 6$, $a = 2$, $j = 5$ |
| 2. $-3a^2 - 4a + 4c$, $a = 2$, $c = 4$ | 4. $3x^2 - x - y + 1$, $x = -8$, $y = -2$ |

5. $-c + 3c - 3k^2 + 2$, $c = 4$, $k = 1$

8. $f(x) = x^2 + 6x - 4$, $f(-9)$

6. $f(x) = 5x^2 + 3x - 5$, $f(6)$

9. $f(x) = 4x^2 - 8x + 2$, $f(-9)$

7. $f(x) = 2x^2 + 9x - 5$, $f(-10)$

10. $f(x) = 6x^2 + 5x + 7$, $f(-4)$

Part D: Distributing and Combining Like Terms

Rewrite each expression in simplest terms. Show the distribution step.

1. $7x + 6 + 4x + 9$

8. $9(11x + 12) + 7x + 4$

2. $9x + 3 - 8x + 4$

9. $11(11x + 7) + 4x + 7$

3. $2x + 5 + x - 3$

10. $9(-6x + 3) + 3x + 2$

4. $9x + 2 + 8x + 7$

11. $-12(-4x + 10) + 12x + 4$

5. $10x + 7x + 11x^2 + 2x^2$

12. $2(8x + 1) + 6x + 5$

6. $12x^2 + x^2 + 2x + 9x$

13. $-7(-4x + 5) - 5x + 10$

7. $3x - 5x^2 + 2x + 9x^2$

14. $-11(-2x + 3) + 6x - 7$

Part E: Absolute Value

Evaluate each expression. Work inside the absolute value bars first.

1. $|4 + 3| - 2| - 3 + 10|$

3. $|2 + 5| - 3| - 12 + 1|$

2. $|-7 - 12| - 3|14 - 10|$

4. $|-13 + 1| - 3| - 14 + 1|$

5. $|10 - 6| + 2|11 + 13|$

8. $|-9 + 7| + 3|5 + 12|$

6. $|-14 - 3| - 3|10 - 13|$

9. $|8 - 10| - 3|-9 + 14|$

7. $|7 + 12| - 3|8 - 4|$

10. $|-3 - 11| + |12 + 11|$

Part F: Fraction Arithmetic

Evaluate. Write every answer in lowest terms.

1. $\frac{1}{5} + 1\frac{1}{2}$

5. $\frac{4}{9} \div \frac{1}{4}$

2. $\frac{2}{3} + 1\frac{1}{2}$

6. $\frac{2}{3} \cdot 1$

3. $2 + 1\frac{5}{6}$

7. $\frac{8}{12}$

4. $\frac{5}{8} \div \frac{1}{6}$

8. $\frac{8}{6}$

Part G: One- and Two-Step Equations

Solve for the variable and simplify your answer.

1. $x + 8 = 16$

5. $9x = 45$

2. $x + 5 = 10$

6. $3x - 10 = -\frac{5}{2}$

3. $x - 6 = -4$

7. $6x - 12 = 60$

4. $x + 1 = -6$

8. $2x + 6 = -1$

11. $9x - 8 = -26$

9. $2x + 5 = 7$

12. $3x - 7 = -22$

10. $7x - 11 = -4$

Part H: Multi-Step Equations with Distribution*Distribute first, then solve for the variable.*

1. $9(x + 1) - 7 = -79$

6. $7(4x + 4) + 11 = 319$

2. $6(7x - 2) + 5 = -385$

7. $6(x - 12) - 2 = -128$

3. $7(7x + 4) - 3 = 74$

8. $2(2x - 2) - 10 = -2$

4. $4(9x - 4) - 1 = -341$

9. $7(5x + 2) - 2 = 362$

5. $3(3x + 11) - 10 = 104$

10. $9(8x - 10) - 5 = 625$

Part I: Variables on Both Sides*Solve for the variable. Clear fractions first where it helps.*

1. $4x + 12 = -7x + \frac{37}{4}$

3. $6x + 1 = -11x - 33$

2. $-2x + 5 = -8x - 37$

4. $x + 10 = -11x + 38$

5. $-2x - 11 = -3x - 2$

11. $\frac{x - 15}{2} - 15 = -18$

6. $2x - 11 = 12x - \frac{43}{3}$

12. $\frac{x + 3}{6} - 2 = -\frac{7}{8}$

7. $-x - 9 = 3x - 53$

13. $-x + 1 = -17$

8. $2x + 9 = 5x + 1$

14. $\frac{x - 5}{4} + 4 = \frac{71}{24}$

9. $-3x - 8 = 9x + 136$

15. $-\frac{7}{3}x + 8 = 43$

10. $8x - 6 = 6x - 22$

16. $\frac{x + 1}{4} + 1 = \frac{13}{4}$

Answer Key

Part A: Numerical Expressions and Order of Operations

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|----------|----------|------------|
| 1. 65 | 5. 52 | 9. -9 |
| 2. 6 | 6. 4 | 10. -144 |
| 3. -12 | 7. -85 | |
| 4. 3 | 8. 77 | |

Part B: Properties of Addition and Multiplication

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|--|---|---|
| 1. A) Commutative Property of Addition | 5. D) Identity Property of Addition | 8. B) Inverse Property of Multiplication |
| 2. C) Commutative Property of Multiplication | 6. D) Identity Property of Multiplication | 9. C) Associative Property of Addition |
| 3. C) Associative Property of Multiplication | 7. B) Inverse Property of Multiplication | 10. C) Associative Property of Multiplication |
| 4. D) Associative Property of Multiplication | | |

Part C: Evaluating Expressions by Substitution

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|----------|--------|--------|
| 1. -27 | 5. 7 | 9. 398 |
| 2. -4 | 6. 193 | 10. 83 |
| 3. 72 | 7. 105 | |
| 4. 203 | 8. 23 | |

Part D: Distributing and Combining Like Terms

- | | | |
|------------------|------------------|-----------------|
| 1. $11x + 15$ | 6. $13x^2 + 11x$ | 11. $60x - 116$ |
| 2. $x + 7$ | 7. $4x^2 + 5x$ | 12. $22x + 7$ |
| 3. $3x + 2$ | 8. $106x + 112$ | 13. $23x - 25$ |
| 4. $17x + 9$ | 9. $125x + 84$ | 14. $28x - 40$ |
| 5. $13x^2 + 17x$ | 10. $29 - 51x$ | |

Part E: Absolute Value

- | | | |
|----------|-------|----------|
| 1. -7 | 5. 52 | 9. -13 |
| 2. 7 | 6. 8 | 10. 37 |
| 3. -26 | 7. 7 | |
| 4. -27 | 8. 53 | |

Part F: Fraction Arithmetic

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|--------------------|-------------------|-------------------|
| 1. $1\frac{7}{10}$ | 4. $3\frac{3}{4}$ | 7. $\frac{2}{3}$ |
| 2. $2\frac{1}{6}$ | 5. $1\frac{7}{9}$ | 8. $1\frac{1}{3}$ |
| 3. $3\frac{5}{6}$ | 6. $\frac{2}{3}$ | |

Part G: One- and Two-Step Equations

1. $x = 8$
2. $x = 5$
3. $x = 2$
4. $x = -7$

5. $x = 5$
6. $x = \frac{5}{2}$
7. $x = 12$
8. $x = -\frac{7}{2}$

9. $x = 1$
10. $x = 1$
11. $x = -2$
12. $x = -5$

Part H: Multi-Step Equations with Distribution

1. $x = -9$
2. $x = -9$
3. $x = 1$
4. $x = -9$

5. $x = 9$
6. $x = 10$
7. $x = -9$
8. $x = 3$

9. $x = 10$
10. $x = 10$

Part I: Variables on Both Sides

1. $x = -\frac{1}{4}$
2. $x = -7$
3. $x = -2$
4. $x = \frac{7}{3}$
5. $x = 9$
6. $x = \frac{1}{3}$

7. $x = 11$
8. $x = \frac{8}{3}$
9. $x = -12$
10. $x = -8$
11. $x = 9$
12. $x = \frac{15}{4}$

13. $x = 18$
14. $x = \frac{5}{6}$
15. $x = -15$
16. $x = 8$